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Paleomagnetism of the Upper Proterozoic and Devonian rocks from the Kłodzko Metamorphic Complex in the West Sudetes (SW Poland): tectonic implications for the Variscan belt of Central Europe

Magdalena Kądziałko-Hofmokl[a,*], Jadwiga Kruczyk[a], Stanisław Mazur[b], Jacek Siemiątkowski[c]

[a] *Institute of Geophysics, Polish Academy of Sciences, 01-452 Warsaw, Ks.Janusza 64, Poland*
[b] *Institute of Geological Sciences, Wrocław University, 50-204 Wrocław, pl.M.Borna 38, Poland*
[c] *Polish Institute of Geology, 53-122 Wrocław, al. Jaworowa 19, Poland*

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Abstract

The Kłodzko Metamorphic Complex (KMC) consists of Upper Proterozoic metaigneous and metasedimentary rocks forming a stack pile thrusted over the Givetian and overlain by Frasnian–Fammenian sediments. Magnetomineralogical experiments show that the magnetic minerals are secondary. The paleomagnetic experiments identified three components of the Natural Remanent Magnetization; labeled A1, A2 and M. The mean pole positions calculated in situ correspond with the Baltica Upper Devonian (A1: PlatS = – 18°, PlongE = 317°), Permo-Carboniferous (A2: PlatS = – 39°, PlongE = 2°) and Triassic–Jurassic (M: PlatS = – 60°C, PlongE = 308°) segments of the Apparent Polar Wander Path (APWP) for Baltica. This indicates that the region studied was situated close to the Baltica plate at least since the Upper Devonian and was not folded after this period.
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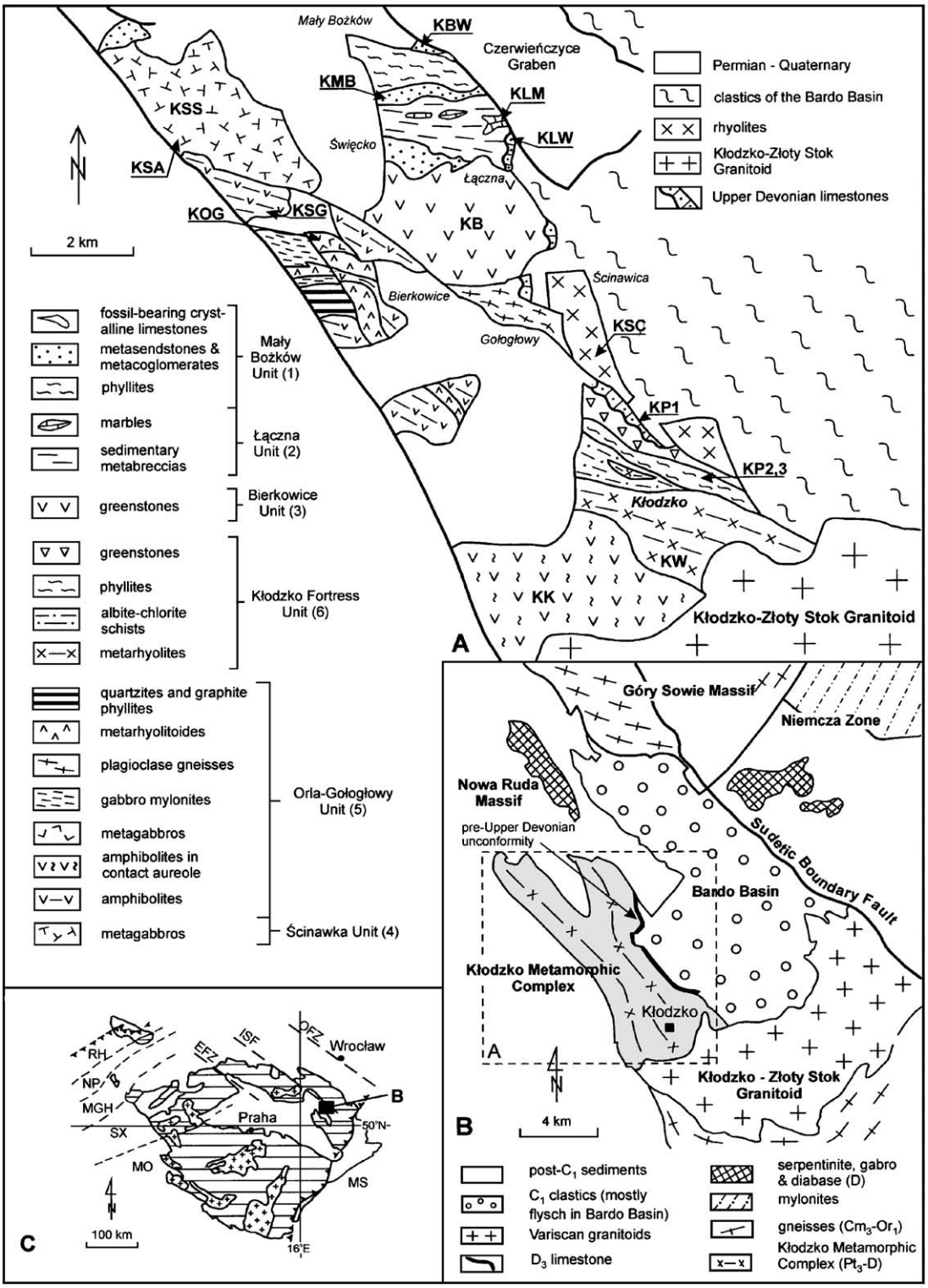
Keywords: Magnetic minerals; Paleozoic; Paleomagnetic pole positions; Remagnetizations; Variscan Belt

1. Introduction

One of key fragments of the Variscan belt of Central Europe, of poorly constrained geological structure and evolution, is the Sudetes area in southwestern Poland and northern Bohemia. Geologically, the area lies on the NE margin of the Bohemian Massif and extends between the WNW and ESE trending Middle Odra Fault Zone in the NE and the parallel Elbe Fault Zone in the SW (Fig. 1, inset C). The Sudetes exposes strongly deformed and metamorphosed complexes of the Variscan internides which define a composite tectonic collage finally assembled during the Carboniferous. The West Sudetes comprises a complex mosaic of geologically distinct fault-bounded pre-Permian units. All basement units in the West Sudetes seem to show some common features indicating their provenance from the Armorican Terrane Assemblage (Tait et al., 2000). One

* Corresponding author.
E-mail addresses: magdahof@igf.edu.pl (M. Kądziałko-Hofmokl), smazur@ing.uni.wroc.pl (S. Mazur).

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Mały Bożków
KBW
Czerwieńczyce Graben
KMB
KLM
KLW
KSS
KSA
Święcko
Łączna
KOG
KSG
KB
2 km
Bierkowice
Ścinawica
KSC
Gołogłowy
KP1
KP2,3
Kłodzko
KW
KK
Kłodzko-Złoty Stok Granitoid
A
Permian - Quaternary
clastics of the Bardo Basin
rhyolites
Kłodzko-Złoty Stok Granitoid
Upper Devonian limestones
fossil-bearing crystalline limestones
metasendstones & metacoglomerates
phyllites
Mały Bożków Unit (1)
marbles
sedimentary metabreccias
Łączna Unit (2)
greenstones
Bierkowice Unit (3)
greenstones
phyllites
albite-chlorite schists
metarhyolites
Kłodzko Fortress Unit (6)
quartzites and graphite phyllites
metarhyolitoides
plagioclase gneisses
gabbro mylonites
metagabbros
amphibolites in contact aureole
amphibolites
Orla-Gologłowy Unit (5)
metagabbros
Ścinawka Unit (4)
Góry Sowie Massif
Niemcza Zone
Nowa Ruda Massif
pre-Upper Devonian unconformity
Sudetic Boundary Fault
Bardo Basin
Kłodzko Metamorphic Complex
Kłodzko
A
Kłodzko - Złoty Stok Granitoid
B
4 km
post-C1 sediments
C1 clastics (mostly flysch in Bardo Basin)
Variscan granitoids
D3 limestone
serpentinite, gabro & diabase (D)
mylonites
gneisses (Cm3-Or1)
Kłodzko Metamorphic Complex (Pt3-D)
RH
NP
MGH
SX
MO
EFZ
ISF
OFZ
Wrocław
Praha
B
50°N
MS
16°E
100 km
C

of these units, the Kłodzko Metamorphic Complex (KMC) is a composite domain that displays fragments of a Neoproterozoic active margin, telescoped during Variscan nappe tectonics, with rift-related Paleozoic sequences (Kryza et al., 2003).

The significance of paleomagnetic data from the metamorphosed pre-Permian basement of the Sudetes has been lively discussed. The studies generally revealed the palaeomagnetic poles representing the Carboniferous and Permian, as well as Triassic–Tertiary overprints (e.g. Jeleńska et al., 1995; Edel et al., 1997; Kądziałko-Hofmokl and El-Hemaly, 1997; Kądziałko-Hofmokl et al., 1999; Jeleńska et al., 2001, 2003). Some results suggest the presence of characteristic directions with Late Devonian poles (Jeleńska et al., 2001, 2003) and poles located at the Early Palaeozoic segment of the Apparent Polar Wander Path (APWP, Torsvik and Smethurst, 1994) for Baltica (Jeleńska et al., 1995; Nawrocki and Żelaźniewicz, 1996). The latter authors pointed out that their Lower Palaeozoic pole agrees with the Avalonia reference path after tectonic correction. The value of such paleomagnetic data for tectonic interpretations was a matter of debate due to the lack of well-established APWPs for components of the Armorican Terrane Assemblage and, sometimes, the not fully demonstrated influence of later remagnetization. The present paleomagnetic study, of the Kłodzko Metamorphic Complex (KMC), contributes to this discussion.

2. Geological setting

The Kłodzko Metamorphic Complex (KMC), geographic latitude 50.5°N, geographic longitude 16.7°E, crops out in the Central Sudetes between the Góry Sowie Massif and the Kłodzko–Złoty Stok Granitoid (Fig. 1). It comprises metasedimentary and various, mostly basic, metaigneous rocks and orthogneisses metamorphosed under greenschist to amphibolite facies conditions (Wojciechowska, 1990; Kryza and Mazur, 2001). These crystalline rocks form a NW–SE elongated outcrop (ca. 100 km^2) bounded in the south and west by Upper Carboniferous to Lower Permian clastic sedimentary sequence of the Intra-Sudetic Basin. In the NE, the KMC is unconformably overlain by the unmetamorphosed Upper Devonian conglomerate and limestone (Bederke, 1924), which form the basal part of the adjacent Bardo Succession. The SE boundary of the KMC is defined by an intrusive contact with the Kłodzko–Złoty Stok Granitoid of uncertain, probably Early Carboniferous age.

The KMC comprises a number of separate tectonic units (Figs. 1 and 2) and is built up, from base to top, by: (1) the Mały Bożków Unit comprising Givetian shelf sequence (Hladil et al., 1999), (2) a mélange body defining the Łączna Unit, (3) the Bierkowice Unit composed of mafic volcanics, (4) metagabbros of the Ścinawka Unit, (5) metagabbros and mafic volcanics of the Orla-Gołogłowy Unit, intruded by granitoids and their subvolcanic equivalents, and accompanied by deep marine sediments (6) and at the very top, distal flysch with basaltic lavas, associated with volcaniclasticic sandstones and dacitic/andesitic tuffs composing the Kłodzko-Fortress Unit. These tectonic elements, interpreted as thrust sheets, reveal contrasting metamorphic paths (Kryza and Mazur, 2001), differences in lithological composition and the occurrence of tectonic contacts. The three lower tectonic units of the KMC, exposed in the NE part of the area, contain metabasic rocks of within-plate tholeiites geochemical characteristics. In contrast, metaigneous rocks from the three upper units show geochemical signature suggesting their supra-subduction affinities (Kryza et al., 2003). The various deviations of protoliths is in accord with their contrasting ages. The lowermost Mały Bożków Unit contains fossil-bearing limestone with the Middle Devonian coralline fauna. On the other hand, the U/Pb zircon dating from the two uppermost units indicates their Upper Proterozoic age in the range of 590–600 Ma (Mazur et al., 2003).

The increasingly higher tectonic units are consecutively exposed from the west to east. The metamorphic

Fig. 1. Geological sketch map of the Kłodzko Metamorphic Complex (A). Inset (B): Geological setting of the Kłodzko Metamorphic Complex within the Góry Sowie–Kłodzko region of Central Sudetes; Inset (C): geological setting of the Kłodzko Metamorphic Complex (denoted by black rectangle) in the Bohemian Massif. EFZ, Elbe Fault Zone; ISF, Intra-Sudetic Fault; MGH, Mid-German Crystalline High; MO, Moldanubian Zone; MS, Moravo–Silesian Zone; NP, Northern Phyllite Zone; OFZ, Odra Fault Zone; RH, Rhenohercynian Zone; SX, Saxothuringian Zone. The letters denote the sampled lithotypes as in Table 1.

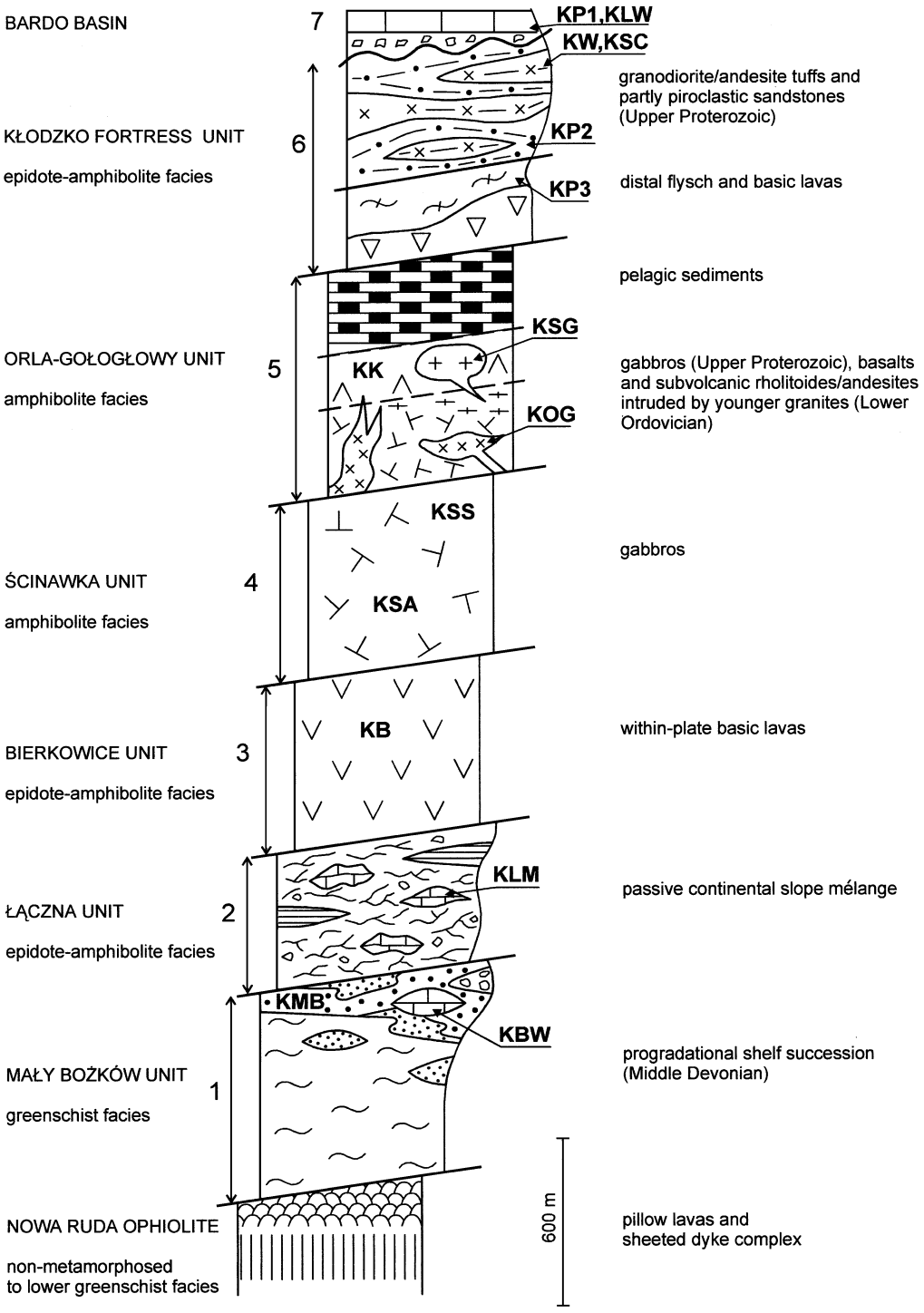


Fig. 2. Lithotectonic log of the Kłodzko Metamorphic Complex. The numbers denote studied units; letters, sampled lithotypes, as in Table 1.

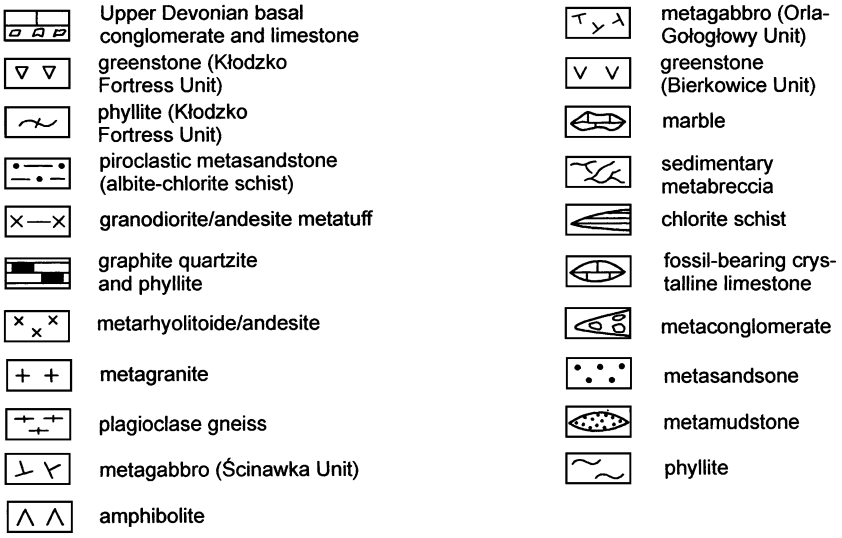


Fig. 2 (*continued*).

grade increases up-profile, from greenschist to amphibolite facies, except for the highest unit that shows epidote–amphibolite facies metamorphism. The major deformation event, a top-to-WNW thrusting, resulted in nappe formation in the KMC.

From the characterised units, 134 hand samples oriented with a geological compass were collected in the KMC and Upper Devonian limestones (Unit 7), from which the standard cylinders were drilled in the laboratory for paleomagnetic and rock-magnetic study.

3. Magnetic mineralogy

The study of magnetic minerals was performed by means of several magnetic and non-magnetic methods in order to identify magnetic carriers of the isolated components of natural remanence.

3.1. Magnetic methods

Four thermomagnetic methods were applied: (a) the thermal decay of isothermal remanence IRM acquired in high fields (1–3 T) during heating to 700 °C in the non-magnetic space, applied to all samples (device by TUS-Warsaw, Kądziałko-Hofmokl and Kruczyk, 1976). The IRM-T curves yield values of blocking temperatures Tb of minerals present in the rock, (b) the three-axis Lowrie method, Lowrie (1990) using three mutually perpendicular fields of: 3, 0.4 and 0.1 T, applied to 36 samples. Three thermal decay curves obtained in the course of the experiment bring values of blocking temperatures Tb of mineral fractions of different coercivities. Measurements of the IRM (isothermal remanence) acquisition curves accompanied the thermomagnetic Lowrie thermoanalysis. (c) Thermal demagnetisation of magnetization I in an external field of different intensity (from several tens to several hundreds of mT) during heating to 600 °C with the Variable Field Translation Balance at the University of Liverpool, applied to 10 samples. The $I–T$ curves yield values of Curie temperatures, T_c, of magnetic minerals. (d) Study of the changes of bulk low field magnetic susceptibility K_b during heating at the low (− 196 °C room temperature) and high (room temperature − 700 °C) temperature ranges, with the susceptibility bridge KLY-3/CS-3 by Agico-Brno, applied to seven samples. Method yields temperatures of low-temperature transitions in its low-temperatures branch and T_c values in its high temperature branch (Orlicky, 1994).

Hysteresis parameters (saturation magnetization, M_s; saturation remanence, M_r; coercivity, H_c; and remanence coercivity, H_{cr}), with the Molspin Vibrating Sample Magnetometer (VSM) of with the highest field of experiment of 1 T, applied to all samples. The values of ranges of H_c are summarized in Table 1.

Table 1
Some characteristics of the studied rocks

Unit name label/*N*	Lithology	Age (Ma)	*I*(NRM) (mA/m)	K_m, (SI × 10^{-3})	H_c (mT)
Upper Devonian **7**/13		370–360,			
KP1	limestone	Bederke	0.16–1.09	0.05–0.16	16–32
KLW	limestone	(1924)	0.2–0.7	0.12–0.21	6–13
Kłodzko–Twierdza **6**/29		590–600,			
KP2,3	schist, phyllite	Mazur et al.	0.14–37	0.5–0.8	1–27
KW	rhyolite	(2003)	0.2–1	0.06–0.125	2–30
KSC	rhyolite		0.2–0.75	0.06–0.09	6–22
Orla-Gołogłowy **5**/31					
KK	amphibolite		0.8–410	0.70–410	1–15
KSG	metagranite			0.15–0.40	2–30
KOG	gabbro		0.03–0.36	0.25–0.85	1–60
Ścinawka **4**/18					
KSA	gabbro			0.14–0.60	1–14
KSS	gabbro–amphibolite		0.3–0.8	0.5–0.7	1–90
Bierkowice **3**/10 KB	metabasite		1–1000	1–70	5–170
Łączna **2**/13	metasandstone, schist (melange metamorphosed		0.015–0.25	−0.006–0.060	3–17
KLM	in greenstone facies)		0.20–0.40	0.3–0.6	1?
Mały Bożków **1**/20		380–370,			
KBW	limestone	Hladil et al.	0.03–0.40	−0.004–0.012	8–100
KMB	metasandstone	(1999)	0.10–0.45	0.150–0.270	0.1?

Labels of the units are shown in bold numbers; *N*, number of hand samples; *I*(NRM), intensity of the natural remanence NRM; K_m, mean susceptibility; H_c, coercivity.

Non-magnetic methods consisted of (a) reflective microscopy applied to 27 samples, (b) scanning electron microscopy (SEM) LEO 1430 with microprobe ISIS 300 by Oxford Instruments (by E. Starnawska, Polish Geological Institute of Warsaw) applied to 17 samples, (c) X-ray analysis performed for three samples where the SEM and microprobe results revealed the presence of different ilmeno-hematite phases.

The use of so many experimental methods enabled the identification of magnetic minerals even in a very weakly magnetic rocks. Only data directly related to the paleomagnetic results are presented here. The magnetic minerals identified in respective units by the methods described above are depicted in Table 2 together with numbers of corresponding figures: Fig. 3a–c presents results of the Lowrie method, Fig. 4a–c results of thermal decay of IRM.

Table 2
Magnetic minerals identified by thermomagnetic methods

Label of the unit/ label of lithology	Magnetic minerals after thermomagnetic and hysteresis study. Number of respective figure
1/KBW + KMB	goethite, pyrrhotite, fine-grained Fe-oxides: magnetite, hematite, Fig. 3a
2/KLM	goethite, maghemite, hematite, small amount of pyrrhotite, Fig. 3c
3/KB	magnetite often with phase of Tub 150–200 °C, Figs. 4a and 5a
4/KSA, KSS	magnetite often with phase of Tub 150–200 °C or only hematite, Figs. 4b and 5b
5/KOG, KK, KSG	magnetite often with phase of Tub 150–200 °C, Fig. 5a
6/KP2, KP3, KW, KSC	magnetite, pyrrhotite, goethite, hematite
7/KLW, KP1	goethite, pyrrhotite, fine-grained Fe-oxides: magnetite/maghemite Figs. 3b and 4c

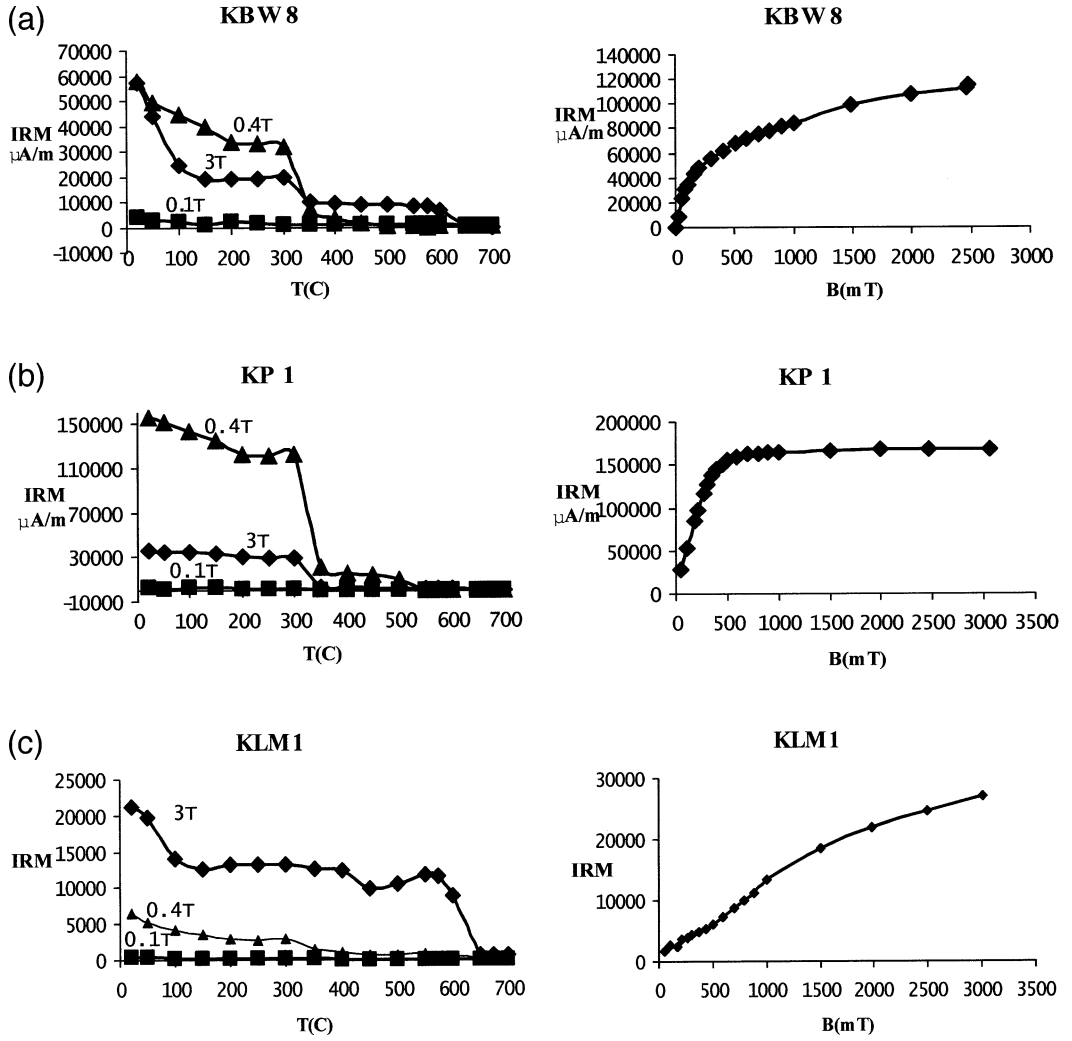


Fig. 3. Complex three-axis analysis according to the Lowrie (1990) method (left) and IRM acquisition curves (right) for: KBW limestone from the unit 1 (a), KP limestone from the unit 7 (b) and KLM metasandstone from the unit 2 (c). NRM, value of isothermal remanence in arbitrary units; *T*, temperature in °C; *B*, magnetizing field in mT. Squares—component imparted in the 0.1-T field, triangles—component imparted in the 0.4-T field, diamonds—component imparted in the 3-T field.

The phase with Tub of 150–160 °C often appearing in metaigneous rocks (see Table 1), and clearly visible in Fig. 4a, disappears after heating specimens to 160–170 °C. The M_r, H_c and H_{cr} measured during step-wise heating decrease in this temperature range, whereas the M_s increase. The $I–T$ curve measured for one of the KB samples in the field as low as 69 mT reveals a hump in the 150–250 °C temperature range (Kądziałko-Hofmokl and Kruczyk, 2002). Those effects signify the presence of fine magnetite grains with maghemite coating. The oxidation processes leading to formation of these structures take place at low, even ambient temperatures and are considered to be of recent origin, (Van Welzen, 1992; Didenko, 1992; Kądziałko-Hofmokl, 2001).

The curves of susceptibility vs. temperature $K_b–T$ (Fig. 5a,b) reveal presence of the Verweey transition of magnetite with T_v temperature of about − 150 °C and T_c of magnetite of about 580 °C (Fig. 5a) in samples of KB and KK. However, in the KK amphibolites (unit 5)

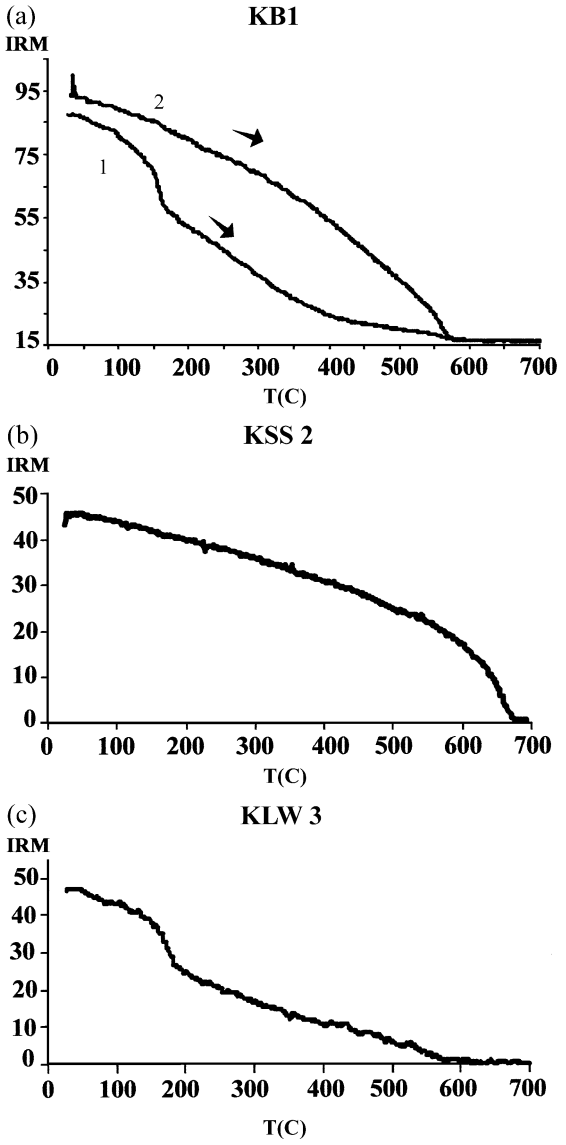


Fig. 4. Curves of thermal decay of the remanence acquired in the field 2.5 T during heating in air in the non-magnetic space (a) for greenstone KB from the unit 3, (b) for amphibolite KSS from the unit 4, (c) for KLW limestone from unit 7. IRM, intensity of remanence in an arbitrary units; *T*, temperature in °C. Curve denoted 1 in (a) represents the first heating to 700 °C, curve denoted 2 represents the second heating of the same specimen. In (b) and (c), only the curves of the first heating are shown.

apart of the T_v, the low-temperature curve reveals the presence of a peak between −180 and 190 °C (Fig. 5b). Similar low-temperature transitions are also present in the KSS gabbro–amphibolites (unit 4), where magnetite was not observed. These peaks represent the T_c of the ilmeno-hematite solid solutions with a bulk composition of about 95% of ilmenite and 5% of hematite (Dunlop and Özdemir, 1997). The ore microscopy and SEM with microprobe performed on KB and KK rocks show presence of magnetite without traces of Ti either in the grains composition or as exsolutions (the lack of Ti in magnetite was supported by the X-ray data). In some cases, magnetite was partly oxidized into hematite in the martite form. Both microscopic methods revealed abundance of ilmenite grains altered partly into titanite and rutile and locally revealing patchy areas of alternating Fe–Ti ratios. The same was proved by microscopic methods in the KSS rocks, where altered ilmenite accompanies needle-like hematite grains. The X-ray analyses performed for the magnetic separates of KB, KK and KSS rocks yielded the characteristics of the solid solutions of ilmeno-hematite series of the same composition as the low-temperature K_b–T data (Lindsley, 1963; Kubicki and Siemiątkowski, 1979). The presence of associations of pure magnetite with no traces of Ti together with the ilmeno-hematite phases containing about 95% of ilmenite serves as a "mineralogical thermometer". According to Spencer and Lindsley (1981), these associations form at temperatures of about 550 °C. This means that the magnetite in the studied KMC rocks is secondary and therefore may carry remanence acquired during or after the thermometamorphic events. Additionally, the microscopic analysis of polished sections proves presence of magnetite grains of post-pyrite origin, therefore originated in still lower temperatures.

The presented results of performed magnetomineralogical study prove that the prevailing magnetic minerals are secondary and that the KMC rocks passed through the same thermometamorphic events.

4. Paleomagnetic experiments and results

The specimens were subjected to standard thermal (TH) and alternating field (AF) demagnetisation, respectively, with a Magnetic Measurements non-magnetic furnace up to 680 °C and 2G Enterprises demagnetizer up to 120 mT. The natural remanence NRM was measured with the 2G Enterprises DC SQUID magnetometer. All equipment was placed within the Magnetic Measurements coil system com-

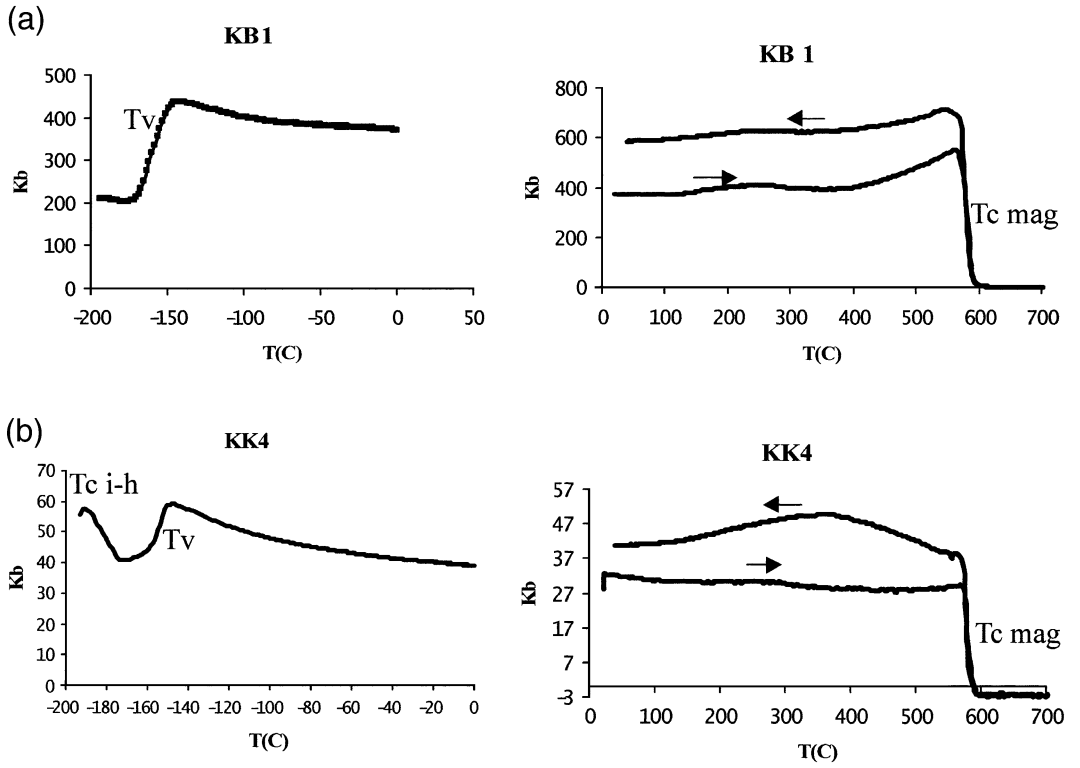


Fig. 5. Bulk susceptibility K_b against temperature in the low-temperature range (left) and high temperature range (right) for the same specimen of greenstone KB as in Fig. 4 (a) and for specimen of amphibolite KK from the unit 5. K_b in arbitrary units, T in °C. T_v: Verweey transition, T_c: Curie temperature, mag: magnetite, i–h: ilmeno-hematite.

pensating 95% of the ambient field. In order to minimise the experimental errors during measurements of weak specimens (intensity of less than 0.4 mA/m^2), the specimen holder was measured before each measurement of NRM of the specimen. The mean susceptibility K_m was measured with the KLY-2/KLY-3 susceptibility bridge of Geofyzika, Brno (now Agico) before thermal cleaning and after each heating step in order to monitor mineralogical changes implied by heating. Ranges of values of K_m and intensity of NRM are included in Table 1.

The results of demagnetizations were analysed, with the special software package (Lewandowski et al., 1997) based on the Kirschvink (1980) principal component analysis and with the method of remagnetization circles of Halls (1976). In some cases, the individual directions were approximated from clusters. Three ranges of unblocking temperatures were isolated: (1) low range LT of 150–200 °C, (2) intermediate range IT of 300–400 °C, and (3) high range HT of Tub>400 °C but only rarely >550 °C. At temperatures >500 °C, the NRM became very noisy and the susceptibility often increased indicating mineralogical changes. The analysis of the AF cleaning revealed presence of low AF components demagnetized in the field of less than 10 mT, high AF components with demagnetization fields >50 mT, and an intermediate range between the two. Thermal cleaning was generally more effective due to the presence of high-coercivity minerals goethite and hematite (see Tables 1 and 2). Mean directions for each unit together with the corresponding pole positions were calculated according to the Fisherian statistics. The results of paleomagnetic analysis are summarized in Tables 3 and 4. All the data are shown in situ because full and partial corrections increased scatter of the isolated directions suggesting that the NRM was acquired after tilting. Two NRM components, M and A, were identified in the three described stability ranges. The majority of directions (100) of components labelled M were of NE (rarely

Table 3
Mean directions, parameters of the Fisher statistics and pole position of the component M for individual units and combined

Unit: name and label	*n*	D/*I*	α_{95}/k	PLatS°/PLongE°	plat°	d*m*/d*p*°
Mały Bożków (KMB, KBW) **1**	18	25/54	9/14	– 66/319	34	9/13
Łączna (KLM) **2**	10	45/49	11/21	– 50/301	30	9/14
Bierkowice (KB) **3**	12	38/48	10/20	– 54/309	29	8/13
Ścinawka (KSA, KSS) **4**	17	23/53	9/17	– 66/323	33	8/12
Orlina–Gołogłowy (KK, KSG, KOG) **5**	12	347/55	10/19	– 52/163	36	10/14
Kłodzko–Twierdza (KW, KSC, KP2,3) **6**	22	38/57	7/22	– 60/297	36	7/10
Upper Devonian (KLW, KP1) **7**	15	338/62	11/13	– 74/90	44	13/17
M combined for the units **1**, **2**, **3**, **4**, **6**	5	34/53	6/139	– 60/308	33	6/9

n, number of directions taken for analysis; *D*/*I*, declination/inclination; α_{95}/k, radius of the confidence circle/precision parameter; Plat/Plong, latitude and longitude of the southern virtual pole position; plat, paleolatitude of the investigated region in the time of magnetization; d*m*/d*p*, half-axes of the ellipses of confidence.

NNW) declinations and positive intermediate (30–60°) or high (>60°) inclinations. (The six directions of the S–SW declinations and negative intermediate or high inclinations were inverted into ones with the NW–NE declinations and positive inclinations for the sake of further analysis.) The M component was found in all temperature ranges. The component A, later divided into A1 and A2 subgroups, comprises directions with the WSW–SES or NW–NE declinations and positive and negative low (<30°) to intermediate inclinations usually not higher than 30°. They appear mainly in IT and HT ranges and in high AF, but some were found also in the LT and low AF. The rarely found directions with the NW–NE declinations were inverted into the SE–SW directions (the results of reversal test performed for A1 and A2 indicate that the confidence circles for normal N and reversed R populations of directions overlap in both cases, see Table 4). The examples of demagnetization diagrams are shown in Fig. 6, the stereographic projections of isolated NRM components directions for each unit of the KMC are presented in Fig. 7.

Comparing the ranges of unblocking temperatures of consecutive isolated NRM components with the results of the magnetomineralogical studies, we may name minerals carrying the respective components.

Table 4
Mean directions, parameters of Fisher statistics and pole positions of components A1 and A2 calculated for each unit and combined

Unit: name and label	Com.	*n*	*D*/*I*	α_{95}/k	PlatS°/PlongE°	plat	d*m*/d*p*
Mały Bożków (KMB, KBW) **1**	A2	12	195/7	11/15	– 41/356	– 4	6/12
Łączna (KLM) **2**							
units **1** and **2** combined	A1	6	233/5	18/10	– 38/0	2	9/18
only unit **2**	A2	6	192/0	25/7	21/318	1	12/25
Ścinawka (KSA, KSS) **4**	A2	12	197/4	12/14	– 38/355	2	6/12
Orlina–Gołogłowy (KK, KSG, KOG) **5**	A1	13	230/3	12/12	– 23/320	0	7/14
Kłodzko–Twierdza (KW, KSC, KP2,3) **6**,							
units **5** and **6** combined	A2	7	187/ – 6	19/11	– 42/7	– 3	10/19
Upper Devonian (KLW, KP1) **7**	A1	4	237/16	16/33	– 13/318	8	9/17
	A2	7	186/10	20/10	– 34/9	5	10/20
A1 combined for units **2**, **5**, **6**, **7**	A1	4	235/8	8/136	– 18/317	4	4/8
A2 combined for units **1**, **2**, **4**, **6**, **7**	A2	5	191/1	8/90	– 39/2	0	4/8
Results of reversal test	A1	10*N*	55/8	11/21			
		20*R*	237/9	7/23			
	A2	16*N*	8/10	11/12			
		28*R*	200/5	8/12			

Com., component; *N*, number of normal directions inverted for further calculations; *R*, number of reversed directions; remaining symbols as in Table 3.

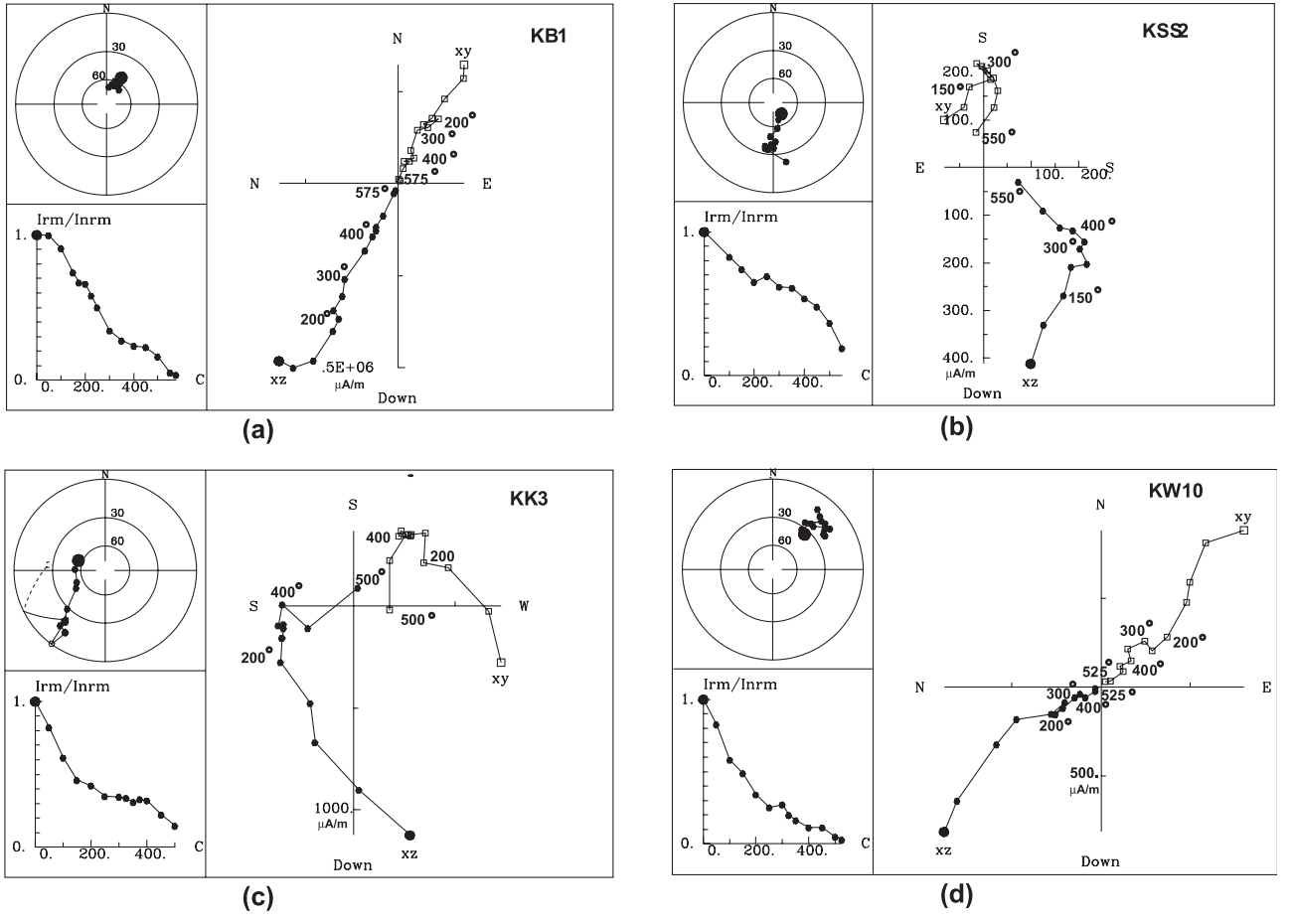


Fig. 6. Examples of the thermal demagnetization results for specimens from greenstones KB of the unit 3 (a), amphibolites KSS from the unit 4 (b), amphibolites KK from the unit 5 (c) and riolites KW from the unit 6 (d). Left up: stereographic projections of the NRM directions, full circles—positive inclinations, open circles—negative inclinations. Left down—decay of the intensity of NRM during the step-like heating. Irm—intensity of NRM after consecutive heating step, Inrm—intensity of the NRM before cleaning. Right—Zijderveld orthogonal projection in two planes: xy—open squares, xz—full circles.

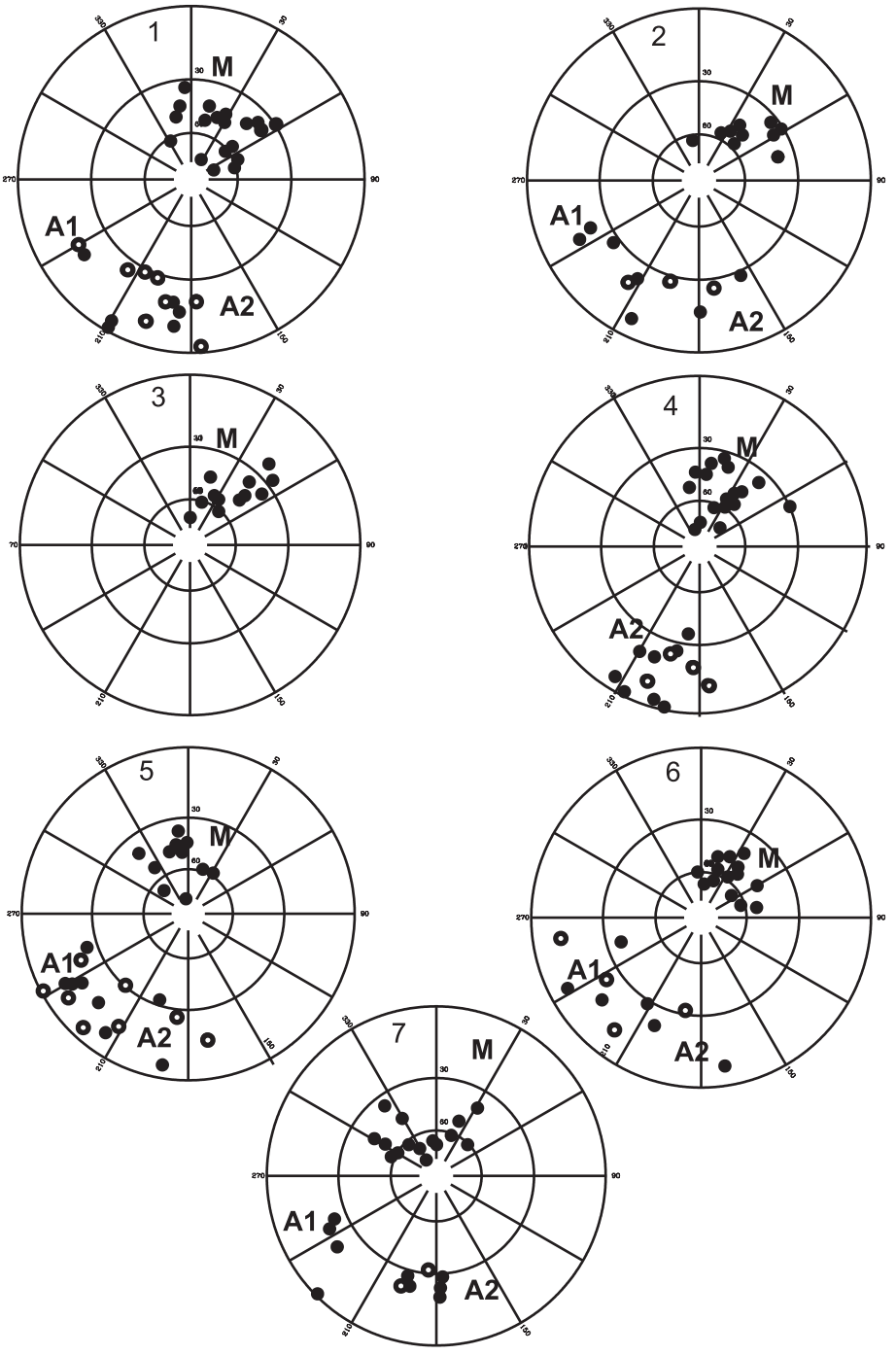


Fig. 7. Stereographic projection of directions of the NRM components isolated due to thermal and AF cleaning in each unit. Numbers denote respective units, M denotes group of directions of component M, A1—group of directions of component A1, A2—group of directions of component A2. Full circles—positive declinations, open circles—negative inclinations.

4.1. Component M

The component M, isolated in the LT range, is either carried by goethite, or is the viscous component carried by Fe-oxides. In the IT range, its carrier is pyrrhotite or maghemite, in the HT range, it is carried by Fe-oxides, mainly magnetite and maghemite (rarely hematite). This component dominates in most units under investigation, such that it was the only component present in unit 3 (in the HT range), and was nearly the only component recognized in unit 6. Table 3 summarizes the mean directions calculated for each unit together with the parameters of Fisherian statistics and corresponding pole positions. Fig. 8 compares the obtained results with the reference Mesozoic and Tertiary data for Stable Europe SE (Besse and Courtillot, 1991) Jurassic of the Krakow–Częstochowa Upland JKC, (Kądziałko-Hofmokl and Kruczyk, 1987) and Tertiary Lower Silesian basalts LS (Kądziałko-Hofmokl et al., 1979). The mean directions of units 1, 2, 3, 4, and 6 form a tight group close to the reference Triassic and Jurassic directions; the corresponding pole positions shown together with their confidence ellipses in Fig. 9a are situated at the Mesozoic segment of the European APWP. The mean direction M (Fig. 8 and Table 3) and the respective mean pole position (Fig. 9b) suggest rather Mesozoic than Tertiary origin of this component. This result indicates that the Sudetes did not move against Stable Europe since the Mesozoic times. Results from units 5 and 7 are shifted away from the remaining ones; they were probably identified not well enough because of changes of mineralogical composition in higher temperatures.

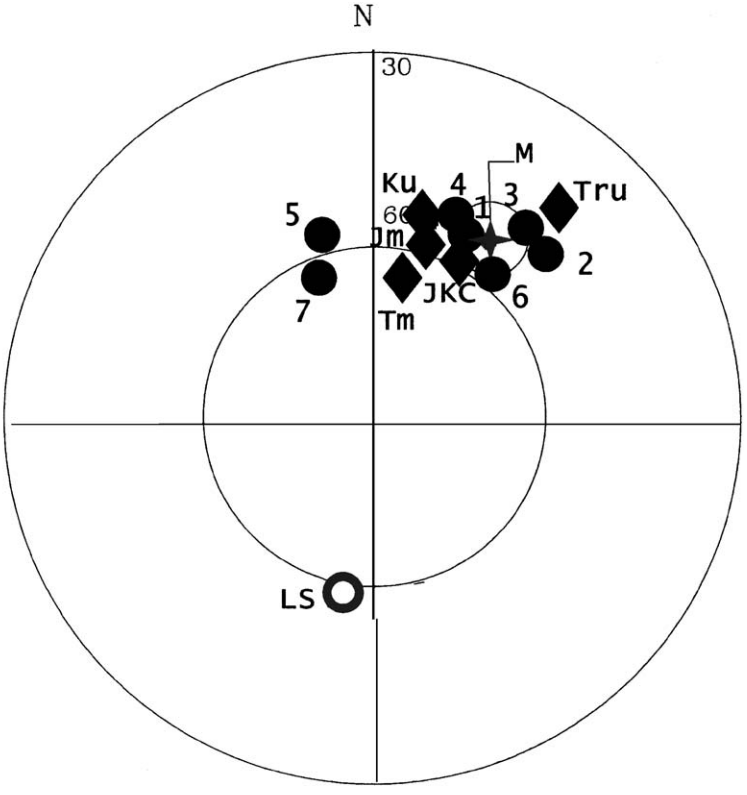
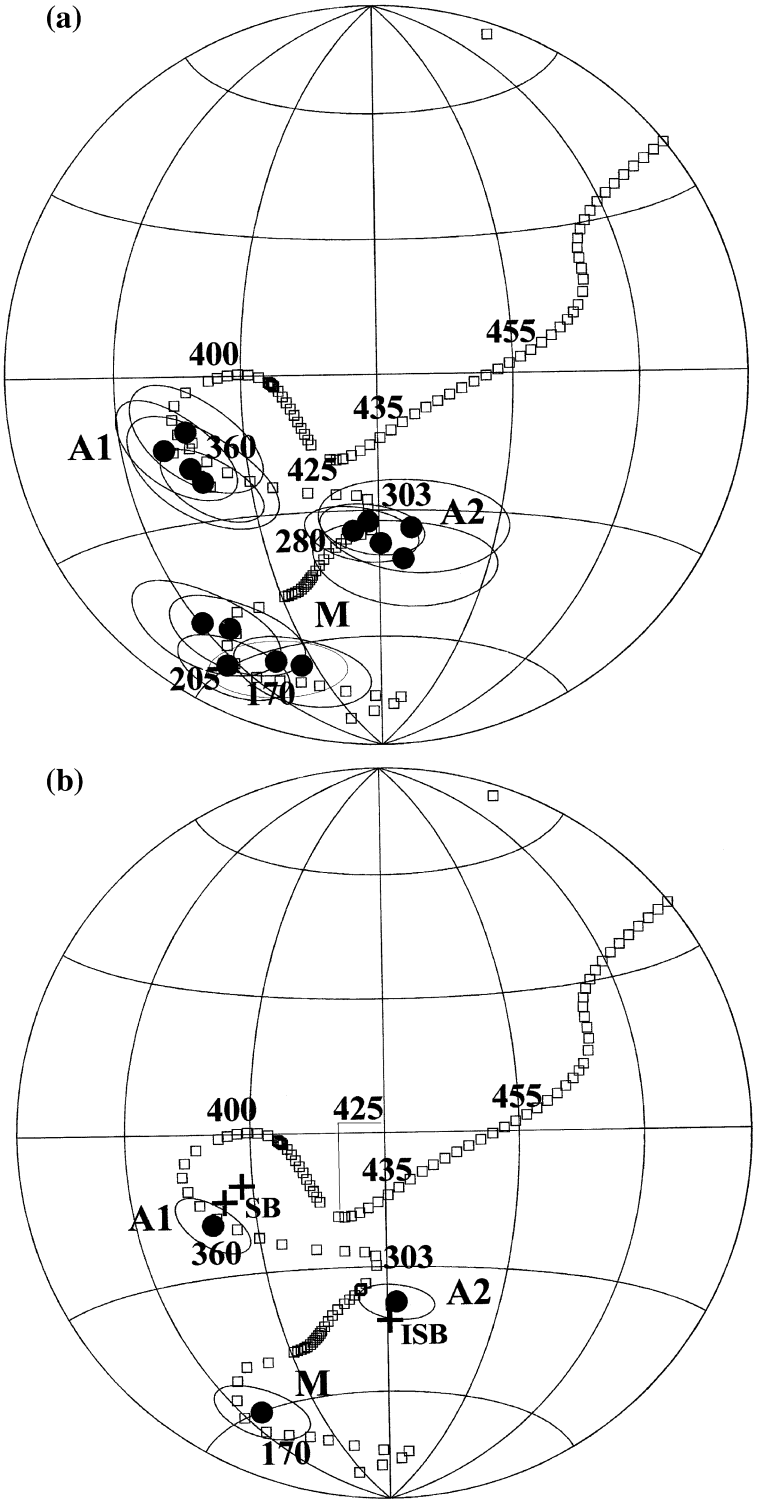


Fig. 8. Stereographic plot of the mean directions of the M component identified in each unit (black circles denoted by respective numbers) together with several reference directions (see Table 3). JKC—Middle–Upper Jurassic data from the Kraków–Częstochowa Upland (Kądziałko-Hofmokl and Kruczyk, 1987). LS—Tertiary Lower Silesia volcanics (Kądziałko-Hofmokl et al., 1979). Star denotes the common mean direction M plotted with its confidence circle. The reference directions calculated after pole positions for Stable Europe of Besse and Courtillot (1991): Tru—Upper Triassic, Jm—Middle/Lower Jurassic, Ku—Upper Cretaceous, Tm—Middle Tertiary.


(a)
400
455
435
A1
360
425
303
A2
280
M
205
170
(b)
400
425
455
435
A1
SB
360
303
A2
ISB
M
170

4.2. Component A

The component A occurs rarely in isolation. It appears most often in the IT and HT Tub ranges where it is carried by pyrrhotite and Fe-oxides: magnetite/maghemite, rarely hematite. Occasionally, the A directions were identified in the LT range. The A component, which occurs in the broad spectrum of declinations, splits into two subgroups A1 and A2. The split is especially clear in units 1 and 7 (Fig. 7). The mean directions of the A1 and A2 subgroups together with their Fisherian parameters and corresponding pole positions for each unit are presented in Table 4 along with the A1 and A2 means combined for the whole KMC.

The A1 and A2 pole positions calculated for each unit with their confidence ellipses are compared with the APWP for Baltica in Fig. 9a. They form two groups situated on the Paleozoic segment of the APW path: the A1 group lies at the Upper Devonian part of the path, the group A2 at the Permo-Carboniferous boundary; the respective combined mean pole positions are plotted in Fig. 9b. For comparison, Fig. 9b presents also the pole positions obtained for the Upper Devonian sediments denoted by crosses-labelled SB (Jeleńska et al., 2001, 2003) and the pole position obtained for the Stephanian–Autunian sediments denoted by star-labelled ISB (Kądziałko-Hofmokl and El-Hemaly, 1997). The A1 pole lies very close to the Upper Devonian SB data indicating that the KMC and the Bardo sedimentary succession simultaneously acquired the Late Devonian remanence and that at that time, they were both close to Baltica. The pole A2 is identical to the Stephanian–Autunian pole position ISB and agrees with the Carboniferous–Permian eastward shift of the Sudetic APWP (Jeleńska et al., 2003).

5. Discussion

The majority of magnetic minerals that carry natural remanence in all studied units are secondary, comprising the following: post-pyrite goethite, pyrrhotite, post-pyrite magnetite, and magnetite formed due to thermometamorphism, post-magnetite maghemite and hematite. The alteration processes took place at temperatures corresponding to the amphibolite-greenstone facies conditions, as well as in temperatures corresponding to hydrothermal alterations and even at ambient temperatures. When the elevated temperatures were higher than the T_c of magnetic minerals, they resulted in acquiring of the new components of NRM, for example, full or partial thermal or thermoviscous remanences, while new magnetic minerals formed at that time may have remanences of chemical or thermochemical origin.

The paleomagnetic data, together with the rock-magnetic experiments, imply that in the KMC rocks, any pre-Upper Devonian paleomagnetic directions were completely obliterated by younger tectonometamorphic events that resulted in mineralogical changes and remagnetizations. The new overprints seem to be acquired during three remagnetization episodes. The first remagnetization event took place at the Latest Devonian probably during deformation and metamorphism of the KMC. The second event was at the Carboniferous–Permian boundary and corresponds to the widespread Permo-Carboniferous overprints in the Sudetes and in the whole of the Variscan belt (see, for example, Edel et al., 1997 and the references within). The third remagnetization phase was related to the Mesozoic uplift of the Sudetes, documented by the hiatus in the sedimentary sequence between the Upper Triassic and the Lower Cretaceous. Although the age of its maximum intensity estimated from the APWP agrees with its Triassic–Jurassic segment, it probably lasted until Tertiary.

6. Conclusions

1. All samples are highly altered, with secondary magnetic minerals: goethite, pyrrhotite, magnetite, maghemite and hematite, often pseudomorphing

Fig. 9. Pole positions obtained in this paper against the Apparent Polar Wander Path for Baltica (Torsvik and Smethurst, 1994). (a) Mean A1 and A2 pole positions obtained for particular units with their confidence ovals (see Table 4) and mean M pole positions for units 1, 2, 3, 4, and 6 with their confidence ovals (Table 3). (b) A1, A2 and M pole positions combined for the whole KMC with corresponding confidence ovals (Tables 3 and 4). SB—pole positions for Upper Devonian sediments from the Bardo Mountains (Jeleńska et al., 2001). ISB—Stephanian–Autunian pole position for sediments from the Intra-Sudetic Basin (Kądziałko-Hofmokl and El-Hemaly, 1997). Numbers along the APWP denote the age in Ma.

other minerals. They carry secondary, post-tilting components of remanence.
2. Natural remanence consists of three overprints acquired in different periods: Upper Devonian (component A1), Carboniferous–Permian boundary (Component A2), Alpine orogeny, particularly Triassic–Jurassic (M).
3. The studied region of the Sudetes was close to the Baltica at least from the Upper Devonian and has not moved against the Stable Europe in the post-Mesozoic times.

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